CHAPTER 4

Factors prevailing for gallstone formation

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Introduction

Gallstone disease is a common pathology of the digestive system with nearly a 10%–20% incidence rate among adults. The prevalence of gallstone disease varies in different populations. The highest prevalence has been found among some North American Indians and Chileans. The lowest prevalence is observed in Asian populations (Barbara et al., 1987). The two components being bile pigments and calcium. Cholesterol and pigment stones are bile pigment and calcium. Various other elements such as iron, phosphorous, carbonate, protein, carbohydrates, mucus, and cellular debris can be found (Cariati, 2015; Sedaghat et al., 1980). Involving calcium bilirubinate as a dominant component, the "pigment stones," which account for 30% of gallstones, are divided into black pigment and calcium bilirubinate stones. Cholesterol gallstones result from the interaction of genetic and environmental factors. These are most likely the consequences of the westernization of modern societies, including a high intake of refined carbohydrates, a high prevalence of obesity, non-insulin-dependent diabetes, and a sedentary lifestyle (Bennion & Grundy, 1978). Furthermore, since biliary cholesterol is mostly derived from the cholesterol produced by plasma lipoproteins, studies have focused on genes related to cholesterol transport. Another group of important risk factors refers to the effects of female hormones that might have an important role in cholesterol metabolism and gallbladder motility (Diehl, 1991).

Gallstone formation occurs as a result of interaction between genetic and environmental factors, of which a few are unmodifiable and unaltered, such as age and genetic makeup. The traditional risk factors for gallstones are four "F": Female; Fertile; Fat; and Forty and one has added a fifth "F"

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for fair skin. When the gallbladder is stable for long periods of time, it continues to secrete mucous, which contributes to slow crystal growth. Black-pigmented gallstones mostly comprise calcium bilirubinate, other calcium salts, and mucoproteins. They are formed in sterile bile in the gallbladder, and unlike cholesterol stones, impaired gallbladder motility appears to play no role in their formation. Insufficient acidification of the gallbladder bile and reduced buffering capacity of mucoproteins also predispose to the development of pigmented stones.

The majority of gallstones are clinically silent and do not develop symptoms; asymptomatic gallstones (biliary pain) have a low rate of complications of approximately 1%–2% per year (Friedman, 1993). This rate is increased in patients with large gallbladders (>3 cm) who develop gallbladder cancers, in patients with sickle cell disease who develop pigment gallstones, in patients with organ transplants, and in patients undergoing bariatric surgery (Ebert et al., 2010). Gallstones may cause no signs or symptoms. If a gallstone lodges in a duct and causes a blockage, the resulting signs and symptoms may occur, which are as follows:

- Sudden and rapidly intensifying pain in the upper right portion of abdomen.
- Sudden and rapidly intensifying pain in the center of abdomen, just below the breastbone.
- Back pain between shoulder blades.
- Pain in right shoulder.
- Nausea and vomiting.
- Gallstone pain may be of several minutes to a few hours.

Who is at the risk of gallstones?

Certain people have a higher risk of developing gallstones than others. Women are more likely to develop than men. Extra estrogen can increase the cholesterol levels in the bile and decrease the gallbladder contractions, which may cause gallstones to form. Women may have extra estrogen due to pregnancy, hormone replacement therapy, or birth control pills. People over 40 are more likely to develop gallstones than younger people, and people with a family history of gallstones have a higher risk.

American Indians have genetic factors that increase the amount of cholesterol in their bile. In fact, American Indians have the highest rates of gallstones in the US; almost 65% of women and 30% of men have gallstones. Mexican Americans are at higher risk of developing gallstones.

Factors underlying gallstone formation

Gallstones, mainly cholesterol gallstones, result from several physiological events such as hepatic secretion of bile saturated with cholesterol, nucleation of cholesterol monohydrate crystals in the gallbladder, and impaired emptying of gallbladder contents (Fig. 4.1). Bile saturation and nucleation show consequences of cholesterol monohydrate crystals in the gallbladder and impaired emptying of gallbladder contents. Bile is an aqueous solution, enriched with water-insoluble hydrophobic lipids such as cholesterol and phospholipids such as dissolved solids, which make up about 3% of hepatic bile. Bile salts are the most prominent solute, averaging 20–30 $mmol^{-1}$ in hepatic bile.

Three types of abnormalities have been considered to be responsible for cholesterol gallstone formation. Cholesterol supersaturation, the essential requirement for cholesterol gall formation, occurs *via* excessive cholesterol biosynthesis, which is the main lithogenic mechanism in obese people. In the nonobese, defective conversion of cholesterol to bile acids occurs due to the low or relatively low activity of cholesterol 7α

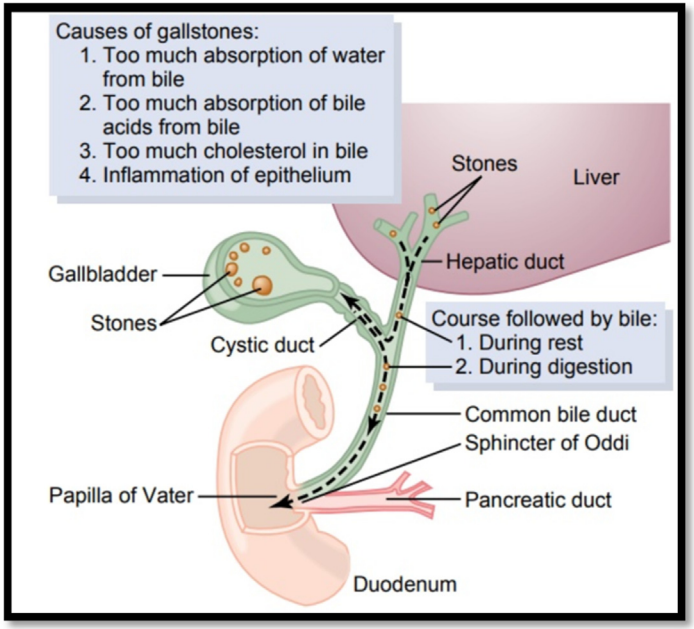

Figure 4.1 Factors underlying gallstone formation figure showing the factors affecting the formation of gallstone.

hydroxylase, the rate-limiting enzyme for bile acid biosynthesis and cholesterol secretion. Impaired gallbladder emptying may result in prolonged stasis of lithogenic bile in the gallbladder, thus allowing more time and a permissive environment for cholesterol crystallization and aggregation of that cholesterol into macroscopic gallstones. A longer stay of bile in the gallbladder may lead to its hyperconcentration by the gallbladder epithelium. The cholesterol saturation index (CSI) of the newly secreted bile is higher due to relatively lower bile salt secretion and relatively higher cholesterol secretion due to large proportions of the bile salt entrapped within the sluggish gallbladder. It promotes cholesterol precipitation (Fridhandler et al., 1983). Some of the significant factors prevailing in gallstone formation are as follows:

- Hepatobiliary disease
- Diet
- Intestinal Hypomotility
- Metabolic syndrome, diabetes, and insulin resistance.

Hepatobiliary disease

People who are obese, especially women, have an increased rate of developing gallstones. Obesity increases the amount of cholesterol in bile, which can cause stone formation. Obesity increases the risk of gallstone formation, probably by increasing fasting and residual gallbladder volumes with resultant bile stasis and increasing cholesterol production and bile cholesterol crystal nucleation. Individuals who gradually reduce to a normal body weight normalize their role of gallstones. By contrast, rapid weight loss increases the risk of stone formation during the period of active dieting by as much as 15–25 fold over that of the non-dieting obese population. As the body breaks down fat during prolonged fasting and rapid weight loss, the liver secretes extra cholesterol into bile (Di Ciaula et al., 2019). Rapid weight loss can also prevent the gallbladder from emptying properly. Low-calorie diets and bariatric surgery—surgery that limits the amount of food a person can eat or digest—lead to rapid weight loss and an increased risk of gallstones (Fig. 4.2).

Diet

Research suggests a diet rich in calories and refined carbohydrates and low in fiber increases the risk of gallstones. Refined carbohydrates are grains processed to remove bran and germ, which contain nutrients and fiber

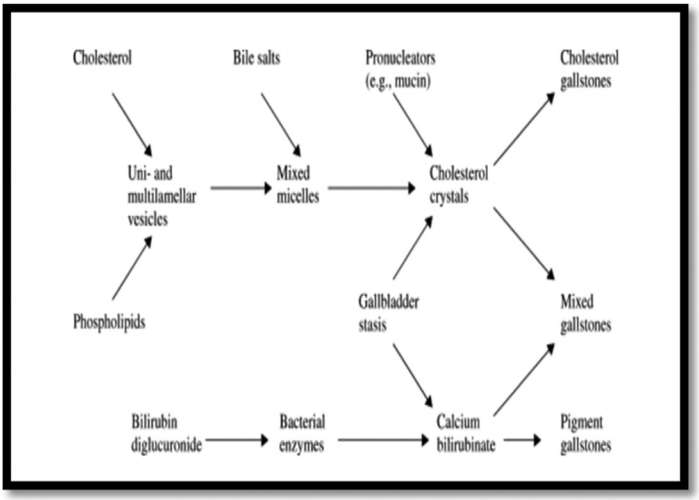

Figure 4.2 Factors for cholesterol gallstone formation and factors prevailing the nucleation of cholesterol crystals. *From Ko, C. W., & Lee, S. P. (2004). Pathophysiology of gallstones. Encyclopedia of Gastroenterology. https://doi.org/10.1016/B0-12-386860-2/00288-4.*

(e.g., refined carbohydrates include white bread and white rice). Increased consumption of meat is associated with gallstone ailment as intake of non-vegetarian food increases biliary acid, a trimethylamine transporter that tempts gallstone cholesterol (Raghu, 2021). A high intake of drenched fats, refined fats, and sugars can possibly amplify the risk of gallstone configuration, while an elevated ingestion of saturated fats and nutritional cholesterol might guard against gallstone formation. The function of dietary factors in cholesterol gallstones has been described by many authors (Sarles et al., 1978, 1970). The risk factors responsible for the development of GSD were food, including meat, cholesterol, tamarind, and foods high in fiber. In cases of such ailments, vegetarian food with suggested calories should be included. A gallstone ailment is caused by the deposit of hard fat or minerals in the gallbladder. In this ailment, the stones block part of the biliary mechanism and create exasperation of the gallbladder and difficulties. This disarray can cause life-intimidating circumstances if left untreated (Huang et al., 2003).

Oils are a significant nutritional factor known to lead to the formation of gallstones. It has been shown that cholesterol, more than bile acids and phospholipids, could be effective in the formation mechanism of gallstones (Shaffer, 2006). The reduction of bile salt and lecithin production in the liver and the increase of cholesterol secretion cause the formation

of gallstones in obese patients. Cholesterol is extremely active in these individuals, and bile is excessively saturated with cholesterol in fasting periods, which results in the formation of gallstones (Li et al., 2009). Cholesterol is a major component of most gallstones and other health ailments, like atherosclerotic plaques. In the cardiovascular system, the excessive accumulation of cholesterol esters in the arterial wall (Tazuma et al., 2017; Thamer, 2022; Wang et al., 2017) leads to clinical atherosclerosis, which mainly occurs in the heart and brain, leading to cardiovascular and cerebrovascular diseases.

Intestinal hypomotility and gallbladder motility

Intestinal hypomotility leads to longer exposure to intestinal microorganisms (bacteria) and the formation of de-conjugated and dehydroxylated bile salts with higher rates than more hydrophobic secondary bile salts. An increased rate of deoxycholate with secondary bile acid is an effective down-regulator of the rate-limiting enzyme in bile acid biosynthesis. Cholesterol hypersecretion to the bile is also potentiated because of increased intestinal absorption.

Among the many factors involved in the pathogenesis of chronic granulomatous disease (CGD), gallbladder hypomotility and the resultant prolonged stasis of lithogenic bile seem to be the most significant. Recently, the discovery of intestinal Cajal-like cells (ICLCs) in the gallbladder provides new insight into the pathogenesis of impaired gall motility (Ahmadi et al., 2010; Pasternak et al., 2012). ICLCs are localized within the muscularis propria. It is believed that ICLCs act as sources of spontaneous electrical slow waves, which are responsible for paced contractions of GBSM, thus regulating the rhythm of gallbladder bile release (Xu et al., 2008). A significantly smaller number of ICLCs was found in the gallbladders of patients with cholelithiasis compared with those without gallstones (Pasternak et al., 2013).

Metabolic syndrome, diabetes, and insulin resistance

These conditions increase the risk of gallstones. Metabolic syndromes also increase the risk of gallstone complications. Metabolic syndrome is a group of traits and medical conditions linked to being overweight or obese that put people at risk for heart disease and type II diabetes. People with diabetes generally have high levels of fatty acids called triglycerides. These increase the risk of gallstones (Chang et al., 2008). Gallbladder function is

impaired in the presence of diabetic neuropathy, and the regulation of hyperglycemia with insulin seems to raise the lithogenic index (Méndez-Sánchez et al., 2005). Pigment stones tend to develop in people who have cirrhosis, a condition in which the liver slowly deteriorates and malfunctions due to chronic or long-lasting injury and infections in the bile ducts. Several hemolytic anemia are conditions in which red blood cells are continuously broken down, such as sickle cell anemia. Gallbladder motility is another risk for cholesterol stones. About 2/3 of patients with cholesterol gallstones display enlarged fasting and postprandial residual gallbladder volume with delayed emptying, which antedates gallstone formation.

A relationship has been reported between gallstones and diabetes, hemolytic anemia, cirrhosis, and ileum resection. High bilirubin levels arising from the destruction of erythrocytes in hemolytic anemia and cholesterol are known to lead to the formation of gallstones (Abdullah et al., 2015). Yıldırım et al. (2008) reported a relationship between pigment stones and liver diseases with a prevalence rising to 30%. Reshetnyak et al. (2009) showed that oral contraceptives and hormone replacement treatment doubled the incidence of cholesterol stones (Reshetnyak et al., 2009; Yıldırım et al., 2008). Estrogen leads to the formation of supersaturated galls by increasing hepatic cholesterol synthesis and reducing bile salt synthesis. The serum progesterone level rises with the use of some oral contraceptives, which can be correlated with inadequate discharge from the gallbladder and increasing residual volume (Reshetnyak, 2012).

A lack of melatonin significantly contributes to gallbladder stones, as melatonin inhibits cholesterol secretion from the gallbladder, enhances the conversion of cholesterol to bile, and is an antioxidant, which can reduce oxidative stress on the gallbladder (Koppisetti et al., 2008). High biliary protein and lipid concentrations are risk factors for the formation of gallstones. A thickened gallbladder mucoprotein known as gallbladder sludge with tiny entrapped cholesterol crystals is the precursor of gallstones (Strasberg et al., 1990). It can cause biliary pain, cholecystitis, or acute pancreatitis (Carey & Cahalane, 1988). The sources of sludge are pregnancy, prolonged total parenteral nutrition, starvation, or rapid weight loss. Pregnancy leads to various changes in gallbladder functions, and gallbladder residual volume increases to double the normal. This explains the increasing frequency of gallstones in fertile women (Durgun, 2000). The biliary calcium concentration plays a role in bilirubin precipitation and gallstone calcification. The use of the somatostatin analog octreotide acts as a contributing factor to the development of gallstones.

Genetic trait

Gallstone formation is a complex genetic trait that involves the effects of susceptible alleles of multiple LITH genes and their interactions with environmental factors. In specific groups of patients, mutations in single genes have been identified that result in a higher risk of gallstone formation. Pullinger et al. (2002) found a link between a single gene defect of cholesterol 7α hydroxylase (CYPTA 1) and cholesterol gallstones associated with hypercholesterolemia resistant to 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors in two homozygote males (Pullinger et al., 2002). Human studies have looked for candidate genes that might favor cholesterol gallstone formation, including expression of apoE (Juvonen et al., 1993; Van Der Linden, 1973), apoB, apoA-1, and cholesterol ester transfer protein. The expression of genes that are involved in insulin resistance and fat deposition is affected by the interactions between genes and environmental factors (Wang et al., 2017).

Ion and water absorption

The major function of the gallbladder is not only to store bile but also to concentrate it by absorbing water and ions. This function of the gallbladder, in conjunction with hepatic hypersecretion of cholesterol, contributes to cholesterol gallstone formation. The fasting gallbladder to hepatic bile ratios for individual and total biliary lipids is a rough estimate of the capacity of gallbladder concentration and are uniformly increased in animals on a lithogenic diet compared with those on a control diet. The removal of water from gallbladder bile involves different epithelial ion transport systems that lead to net NaCl apical absorption (Meyer et al., 2005). In response to a lithogenic diet, ion transport is released from these inhibitory factors, resulting in enhanced gallbladder absorption (Narins et al., 2005). Giurgiu et al. (1997) demonstrated an increased net Na^+ absorption during the early pre-crystal stage of gallstone formation.

The initiating factors that lead to the observed dysfunctional gallbladder ion transport are not clear. Since these changes occur only when cholesterol-supersaturated bile is present in the gallbladder, direct exposure of the gallbladder mucosa to lithogenic bile is highly suspected to be the trigger event.

Medical treatment and surgical treatment are possible gallstone treatments. Medical treatment involves the administration of bile acids to dissolve gallstones, nonsteroid antiinflammatory drugs, and antispasmodic

drugs (Tazuma et al., 2017). Surgical treatment is the standard treatment for gallstone disease. It includes a laparoscopic cholecystectomy and open surgery. The use of open surgery or laparoscopic surgery depends on the sex, age, and gallbladder wall thickness of the patient and the presence of acute cholecystitis (Thamer, 2022).

Conclusions

The basic role of the gallbladder is to enhance the digestive power of bile through a concentrating mechanism, drive the flow of bile salt through the enterohepatic circulation *via* periodic motion, and maintain the thermodynamic stability of gallbladder bile through differential absorption of lipids. These functions of the gallbladder are all impaired and constitute essential contributing factors in CGD. It serves as a link in coordinating metabolic homeostasis, maintenance of normal body composition, and insulin sensitivity. Gallstones, mainly cholesterol gallstones, result from several physiological events such as hepatic secretion of bile saturated with cholesterol, nucleation of cholesterol monohydrate crystals in the gallbladder, and impaired emptying of gallbladder contents. Gallstone formation occurs as a result of interaction between genetic and environmental factors, of which a few are unmodifiable and unaltered, such as age and genetic makeup. A longer stay of bile in the gallbladder may lead to its hyperconcentration by the gallbladder epithelium. The CSI of the newly secreted bile is higher due to relatively lower bile salt secretion and relatively higher cholesterol secretion due to large proportions of the bile salt entrapped within the sluggish gallbladder. The majority of gallstones are clinically silent and do not develop symptoms; asymptomatic gallstones (biliary pain) have a low rate of complications of approximately 1%–2% per year. Current concepts on gallstone formation invoke the notion that nucleation of cholesterol crystals occurs in a mucous gel through protein–lipid interactions. In short, avoiding or managing different risk factors and adjusting diet accordingly set a path towards better gallstone prevention, whether a person is prone to or has ever experienced a painful stone.

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